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Part B

→ lab4 □ python3 lab4.py ALL lab4.pcap www.youtube.com

CASE A:

Frame 1

Frame 215

IPAddr-computer:

IPAddr-gateway: 100.69.32.1

CASE B:

Frame 232

Frame 290

Frame 293

Frame 310

Frame 365

Frame 844

Frame 903

Frame 1055

Frame 1073

IPAddr-gateway: 100.69.32.1

MACAddr-gateway: 00:00:0c:9f:f6:a4

Packet hex (ARP reply frame 293):

```
4a28ed5ec17d0000c9ff6a40806000108000604000200000c9ff6a4644520014a28ed5ec17d64  
4522180000000000000000000000000000000000000000000000000000000000
```

CASE C:

Frame 1322

Frame 1323

Frame 137

Frame 1329

Frame 1330

name: www.youtube.com

IPAddr: 100.69.45.122

Part C

1.

The protocol used to assign the IP address is **DHCP (BOOTP/UDP)**.

The protocol used to obtain the MAC address of the gateway is **ARP**.

The first one to appear in the pcap is **DHCP**, because Frame 17 occurs before Frame 248.

2.

No. The pcap only contains ARP requests/replies for the gateway IP **100.69.32.1**.

There are no ARP replies corresponding to any other hosts.

3.

The DNS response does not include a public CNAME because the DNS query was resolved by a local ISP resolver using CGNAT.

Number of IP addresses found: **1**

Returned IP address: **100.69.42.51**

4.

The connection to www.youtube.com uses **TCP**, because HTTPS/TLS operates over TCP port 443, requiring a reliable stream.

5.

The IP 100.69.42.51 belongs to **CGNAT** (100.64.0.0/10), so it is internal to the ISP and has no public geolocation.

Therefore, it is not possible to determine physical proximity, but it is likely local to the ISP.